

Kubernetes Namespaces – Quick Revision

Why do we use Namespaces?

Used to separate Dev, Test, UAT and Production environments. Resources are isolated between namespaces.

1. List all namespaces

kubectl get ns

Displays all namespaces. Default namespaces: default, kube-system, kube-public, kube-node-lease.

2. List pods in all namespaces

kubectl get pods -A

or **kubectl get pods --all-namespaces**

Shows pods from every namespace.

3. View system pods

kubectl get pods -n kube-system

Shows Kubernetes system components like CoreDNS, kube-proxy, Metrics Server.

Default namespace

If no namespace is specified, resources are created in the **default** namespace.

4. Create a namespace

kubectl create ns dev

or **kubectl create namespace dev**

5. Specify namespace in YAML

Add **metadata.namespace: dev** in the manifest to always create the resource in that namespace.

6. Create resource in a namespace

kubectl create -f pod.yaml -n dev

If YAML has no namespace, this creates it in dev.

7. Check current namespace (kubens)

kubens – Shows current namespace.

kubens test – Switches to test namespace.

8. Check current namespace (kubectl)

kubectl config view --minify | grep namespace

Displays current namespace.

9. Change default namespace

kubectl config set-context --current --namespace=dev

Changes the current context's default namespace.

Important Interview Commands

kubectl get ns – List namespaces

kubectl get pods -A – List all pods

kubectl get pods -n kube-system – System pods

kubectl create ns dev – Create namespace

kubectl create -f pod.yaml -n dev – Create resource in namespace

kubectl config set-context --current --namespace=dev – Change default namespace

kubectl config view --minify | grep namespace – Check current namespace

kubens – Show current namespace

kubens test – Switch namespace

Interview Summary

A namespace is a logical partition in Kubernetes used to isolate and organize resources within the same cluster.

Additional Notes

10. List resources in the current namespace

Examples:

```
kubectl get pods
```

```
kubectl get svc
```

```
kubectl get deploy
```

```
kubectl get all
```

Notes

- Whenever you delete a namespace, all resources inside that namespace are also deleted.
- You can delete the namespace that you are presently in. After it is deleted, commands against that namespace will fail until you switch to another namespace.